Experiment 17: VoteOnline – Online E-Voting System

Aim:

To design and develop an interactive online e-voting system that enables voters to log in using their Voter ID, verify their identity using OTP, select a candidate, submit their vote, and view election results. The system also allows administrators to manage candidates, voter lists, and election results.

Algorithm:

1. Design a responsive interface for voters and administrators using HTML, CSS, and JavaScript.
2. Initialize candidate details, voter records, and election settings using browser local storage.
3. Accept the Voter ID and generate a simulated OTP for login verification.
4. Verify the OTP and check whether the voter has already submitted a vote.
5. Display the candidate list and allow the voter to select and confirm one candidate.
6. Record the vote and dynamically update the election statistics and result charts.
7. Provide an administrator dashboard to add candidates, import voter lists, manage election status, and export results.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>VoteOnline | Digital Election</title>
<style>
:root{--bg:#0b1020;--panel:#141c30;--panel2:#1b2640;--line:#293650;
--text:#eef3ff;--muted:#9baac5;--blue:#7195ff;--green:#36d6a0;--red:#ff7185}
```

*{box-sizing:border-box}

body{margin:0;background:radial-gradient(circle at 10% 0%,#202d55,transparent 35%),var(--bg);

color:var(--text);font-family:Inter,Segoe UI,Arial,sans-serif}

button,input,select{font:inherit}

button{cursor:pointer;transition:.2s}

button:hover{filter:brightness(1.13);transform:translateY(-1px)}

.top{height:75px;padding:0 5%;display:flex;align-items:center;justify-content:space-between;

border-bottom:1px solid var(--line);background:#0b1020eF;position:sticky;top:0;z-index:5}

.brand{font-size:23px;font-weight:800;letter-spacing:-1px}

.brand span{color:var(--blue)}

.pill{border:1px solid #315345;background:#112b2b;color:var(--green);

border-radius:30px;padding:8px 13px;font-size:12px}

.layout{max-width:1250px;margin:30px auto;padding:0 20px;display:grid;

grid-template-columns:220px 1fr;gap:25px}

.side,.panel,.stat,.candidate,.hero{background:linear-gradient(145deg,var(--panel),#11182a);

border:1px solid var(--line);border-radius:18px}

.side{padding:18px;height:max-content;position:sticky;top:95px}

.nav{width:100%;display:flex;gap:10px;align-items:center;text-align:left;

border:0;background:transparent;color:var(--muted);padding:13px;border-radius:10px;margin:3px 0}

.nav.active,.nav:hover{background:#263455;color:white}

main{min-width:0}

.hero{padding:27px;background:linear-gradient(115deg,#263c70,#17243e 60%,#173e43);

display:flex;justify-content:space-between;gap:15px;align-items:center;margin-bottom:22px}

```
.eyebrow{color:#a8bdff;font-size:12px;text-transform:uppercase;letter-spacing:2px;font-weight:700}

h1{font-size:clamp(25px,4vw,36px);margin:10px 0}

h2{font-size:20px;margin:0 0 17px}

p{color:var(--muted);line-height:1.6}

.hero p{margin-bottom:0}

.clock{font-size:22px;font-weight:800;color:#b9caff;text-align:center}

.clock small{display:block;color:#b5c6e6;font-size:10px;letter-spacing:1px;margin-top:6px}

.stats{display:grid;grid-template-columns:repeat(3,1fr);gap:14px;margin-bottom:22px}

.stat{padding:18px}

.stat small{color:var(--muted)}

.num{font-size:29px;font-weight:800;margin-top:12px}

.panel{padding:23px;margin-bottom:22px}

.section-head{display:flex;align-items:center;justify-content:space-between;gap:12px;flex-wrap:wrap}

.candidates{display:grid;grid-template-columns:repeat(2,minmax(0,1fr));gap:15px}

.candidate{padding:18px;transition:.2s}

.candidate.selected{border-color:var(--blue);box-shadow:0 0 0 2px #7195ff30}

.avatar{width:58px;height:58px;border-radius:16px;background:#26375a;

display:grid;place-items:center;font-size:28px;margin-bottom:14px}

.candidate h3{margin:0 0 5px}

.party{font-size:12px;color:var(--muted)}

.description{font-size:13px;min-height:42px}

.btn{border:0;border-radius:10px;background:var(--blue);color:#0b1020;

font-weight:800;padding:12px 16px}

.btn.secondary{background:#293650;color:var(--text);border:1px solid #3b4b69}

.btn.green{background:var(--green)}
```

```
.btn.red{background:var(--red)}
.btn:disabled{opacity:.4;cursor:not-allowed;transform:none}
input,select{width:100%;padding:12px 13px;background:#0c1425;color:var(--text);
border:1px solid #33425e;border-radius:10px;outline:none;margin:7px 0 13px}
input:focus,select:focus{border-color:var(--blue)}
label{font-size:13px;color:#c3cee4}
.formrow{display:grid;grid-template-columns:1fr 1fr;gap:12px}
.notice{padding:13px;border-radius:10px;background:#182943;color:#c7d7ff;
font-size:13px;margin:12px 0;line-height:1.6}
.success{background:#12352d;color:#a7ffdf}
.warning{background:#392b1d;color:#ffdb9c}
.hidden{display:none!important}
.barrow{margin:20px 0}
.barlabel{display:flex;justify-content:space-between;gap:10px;font-size:13px}
.track{height:11px;background:#263149;border-
radius:20px;overflow:hidden;margin-top:9px}
.fill{height:100%;width:0;background:linear-gradient(90deg,#6e8dff,#38d5a0);
border-radius:20px;transition:width .6s}
.tablewrap{overflow-x:auto}
table{width:100%;border-collapse:collapse;font-size:13px}
th,td{text-align:left;padding:13px;border-bottom:1px solid var(--line);white-
space:nowrap}
th{color:#aebfe0}
.badge{border-radius:30px;background:#193b32;color:var(--green);padding:5px
9px;font-size:11px}
footer{text-align:center;padding:30px;color:#71809d;font-size:12px}
.toast{position:fixed;bottom:25px;right:25px;background:#193b32;color:white;pad
ding:15px 20px;
```

border:1px solid #3c806c;border-radius:12px;z-index:20;box-shadow:0 10px 35px #0007}

@media(max-width:760px){.layout{grid-template-columns:1fr;margin-top:16px}.side{position:static;

display:flex;overflow:auto;gap:4px;padding:8px}.nav{white-space:nowrap;width:auto}

.stats{grid-template-columns:1fr}.candidates{grid-template-columns:1fr}.hero{align-items:flex-start}

.clock{display:none}.formrow{grid-template-columns:1fr}}

</style>

</head>

<body>

<header class="top">

** VoteOnline. **

** ● DEMO ELECTION **

</header>

** **

<aside class="side">

<button class="nav active" data-page="home">◈   Overview</button>

<button class="nav" data-page="vote">▣   Cast Your Vote</button>

<button class="nav" data-page="history">◷   My Voting Status</button>

<button class="nav" data-page="results">▥   Live Results</button>

<button class="nav" data-page="admin">⚙   Admin Console</button>

</aside>

<main>

<section class="hero">

** **

```
  <div class="eyebrow">Election management system</div>
  <h1>Every vote matters.</h1>
  <p>A modern, interactive voting experience with transparent demo results.</p>
 </div>
 <div class="clock"><span id="clock">--:--:--</span><small>LOCAL TIME</small></div>
</section>

<div class="stats">
 <div class="stat"><small>Total registered voters</small><div class="num" id="voterCount">0</div></div>
 <div class="stat"><small>Votes submitted</small><div class="num" id="voteCount">0</div></div>
 <div class="stat"><small>Participation rate</small><div class="num" id="turnout">0%</div></div>
</div>

<section id="home" class="page">
 <div class="panel">
  <div class="section-head"><h2>Election overview</h2><span class="badge" id="electionStatus">OPEN</span></div>
  <p id="electionDescription">Choose your candidate after voter verification. Each demo voter ID can vote only once.</p>
  <div class="notice">🔐 <b>Two-step demo verification</b><br>Enter a voter ID, then verify the one-time code shown by this local demonstration.</div>
  <button class="btn" onclick="showPage('vote')">Get Started →</button>
 </div>
 <div class="panel">
  <div class="section-head"><h2>Candidate directory</h2><button class="btn secondary" onclick="showPage('results')">View results</button></div>
```

```
  <div class="candidates" id="homeCandidates"></div>
 </div>
</section>

<section id="vote" class="page hidden">
 <div class="panel">
  <h2>01 / Voter verification</h2>
  <p>Enter a voter ID. For this demo, any non-empty ID is accepted unless already used.</p>
  <label for="voterId">Voter ID</label>
  <input id="voterId" placeholder="e.g. VOTER1001" maxlength="40">
  <button class="btn" id="requestOtp">Request login OTP</button>
  <div id="otpArea" class="hidden">
   <div class="notice warning">Demo OTP: <b id="demoOtp"></b><br>This code is displayed here only for testing.</div>
   <label for="otpInput">Enter OTP</label>
   <input id="otpInput" inputmode="numeric" maxlength="6" placeholder="6-digit code">
   <button class="btn green" id="verifyOtp">Verify and continue</button>
  </div>
  <div id="loginMessage" aria-live="polite"></div>
 </div>

 <div id="ballotArea" class="panel hidden">
  <div class="section-head"><h2>02 / Select your candidate</h2><span class="badge">BALLOT</span></div>
  <p>Select one candidate, review your choice, then confirm your vote.</p>
  <div id="ballotCandidates" class="candidates"></div>
  <div id="selectedMessage" class="notice">No candidate selected yet.</div>
```

```
  <button id="submitVote" class="btn green" disabled>Confirm and submit vote</button>
 </div>
</section>

<section id="history" class="page hidden">
 <div class="panel">
  <h2>My voting status</h2>
  <p>Verify your voter ID above to view its local demo status. Your candidate choice is not shown here.</p>
  <div class="formrow">
   <div><label for="historyId">Voter ID</label><input id="historyId" placeholder="Enter your voter ID"></div>
   <div style="align-self:end"><button class="btn" onclick="checkHistory()">Check status</button></div>
  </div>
  <div id="historyResult" class="notice">No voter ID checked.</div>
 </div>
</section>

<section id="results" class="page hidden">
 <div class="panel">
  <div class="section-head"><h2>Live election results</h2><span class="badge">AUTO-REFRESH</span></div>
  <p>Bars and vote counts update when votes are cast in this browser.</p>
  <div id="resultsBars"></div>
  <div class="notice">This demo displays candidate totals. It is not independently audited or cryptographically secured.</div>
 </div>
</section>
```

```
<section id="admin" class="page hidden">
 <div class="panel" id="adminLogin">
  <h2>Administrator access</h2>
  <p>Sign in to manage candidates and demo voter records.</p>
  <label for="adminPassword">Demo admin password</label>
  <input id="adminPassword" type="password" placeholder="Enter admin password">
  <button class="btn" onclick="loginAdmin()">Unlock console</button>
  <div class="notice">Demo password: <b>admin123</b>. Do not use this pattern for real authentication.</div>
  <div id="adminMessage"></div>
 </div>
 <div id="adminArea" class="hidden">
  <div class="panel">
   <div class="section-head"><h2>Candidate management</h2><button class="btn secondary" onclick="logoutAdmin()">Lock console</button></div>
   <div class="formrow">
    <div><label for="candidateName">Candidate name</label><input id="candidateName" placeholder="Full name"></div>
    <div><label for="candidateParty">Party / group</label><input id="candidateParty" placeholder="Party or group"></div>
   </div>
   <label for="candidateDescription">Candidate description</label>
   <input id="candidateDescription" placeholder="Short candidate introduction">
   <button class="btn" onclick="addCandidate()">+ Add candidate</button>
  </div>
  <div class="panel">
   <h2>Voter list management</h2>
```

```
    <p>Import voter IDs from a CSV or text file. The first column or first value on each line is treated as the voter ID.</p>
    <input id="voterFile" type="file" accept=".csv,.txt">
    <button class="btn secondary" onclick="importVoters()">Import voter list</button>
    <div class="tablewrap" style="margin-top:18px"><table>
     <thead><tr><th>Voter ID</th><th>Status</th><th>Action</th></tr></thead>
     <tbody id="voterTable"></tbody>
    </table></div>
   </div>
   <div class="panel">
    <h2>Election controls</h2>
    <label for="electionToggle">Election status</label>
    <select id="electionToggle" onchange="setElectionStatus(this.value)">
     <option value="open">Open for voting</option>
     <option value="closed">Close election</option>
    </select>
    <button class="btn red" onclick="resetElection()">Reset all demo votes</button>
    <button class="btn secondary" onclick="exportResults()">Export results CSV</button>
    <div class="notice warning">Reset permanently clears local demo voting records in this browser. Export first if you need a copy.</div>
   </div>
  </div>
</section>
</main>
</div>
<footer>VoteOnline · Internet Programming Lab · Educational demonstration only</footer>
```

```
<div id="toast" class="toast hidden" role="status"></div>


<script>

const KEY="voteonline_advanced_demo_v1";

const initial={

 candidates:[

  {id:1,name:"Aarav Sharma",party:"Progress Alliance",description:"Focused on education, digital access and public services.",icon:" 👨‍💼 "},

  {id:2,name:"Diya Nair",party:"People First",description:"Prioritizes sustainability, inclusion and community growth.",icon:" 👩‍💼 "},

  {id:3,name:"Kabir Rao",party:"Future Forum",description:"Supports innovation, skills development and infrastructure.",icon:" 👨‍💼 "},

  {id:4,name:"Meera Iyer",party:"Unity Group",description:"Advocates transparent governance and local development.",icon:" 👨‍💻 "}

 ],

 voters:[],

 votes:[],

 open:true,

 nextId:5

};

let db=loadDB(), currentVoter="", pendingOtp="", selectedCandidate=null, verified=false, adminUnlocked=false;


function loadDB(){

 try{

  const saved=JSON.parse(localStorage.getItem(KEY));

  if(saved && Array.isArray(saved.candidates) && Array.isArray(saved.voters) && Array.isArray(saved.votes))

    return saved;
```

```
 }catch(e){}
 return structuredClone(initial);
}
function saveDB(){localStorage.setItem(KEY,JSON.stringify(db));renderAll()}
function escapeHTML(s){return
String(s).replace(/[&<>"']/g,c=>({"&":"&amp;","<":"&lt;",">":"&gt;","'":"&quot;","'":"&#
39;"}[c]))}
function toast(msg){
 const
el=document.getElementById("toast");el.textContent=msg;el.classList.remove("hi
dden");
 setTimeout(()=>el.classList.add("hidden"),2600);
}
function showPage(id){

document.querySelectorAll(".page").forEach(p=>p.classList.toggle("hidden",p.id!=
=id));

document.querySelectorAll(".nav").forEach(b=>b.classList.toggle("active",b.datase
t.page===id));
 if(id==="admin" && !adminUnlocked)
document.getElementById("adminPassword").focus();
 if(id==="results")renderResults();
 window.scrollTo({top:0,behavior:"smooth"});
}
document.querySelectorAll(".nav").forEach(b=>b.addEventListener("click",()=>sho
wPage(b.dataset.page)));

function candidateCard(c,ballot=false){
 const chosen=selectedCandidate===c.id;
 return `<article class="candidate ${chosen&&ballot?"selected":""}">
```

```
  <div class="avatar">${c.icon||"🗳️"}</div>
  <h3>${escapeHTML(c.name)}</h3><div class="party">${escapeHTML(c.party)}</div>
  <p class="description">${escapeHTML(c.description||"Candidate information")}</p>
  ${ballot?`<button class="btn ${chosen?"green":"secondary"}" onclick="selectCandidate(${c.id})">${chosen?"✓ Selected":"Select candidate"}</button>`:""}
 </article>`;
}
function renderCandidates(){

document.getElementById("homeCandidates").innerHTML=db.candidates.map(c=>candidateCard(c)).join("")||"<p>No candidates added.</p>";

document.getElementById("ballotCandidates").innerHTML=db.candidates.map(c=>candidateCard(c,true)).join("")||"<p>No candidates available.</p>";
}
function renderResults(){
 const total=db.votes.length;
 document.getElementById("resultsBars").innerHTML=db.candidates.map(c=>{
  const count=db.votes.filter(v=>v.candidateId===c.id).length;
  const percent=total?Math.round(count/total*100):0;
  return `<div class="barrow"><div class="barlabel"><b>${escapeHTML(c.name)}</b><span>${count} votes · ${percent}%</span></div>
  <div class="track"><div class="fill" style="width:${percent}%"></div></div></div>`;
 }).join("")||"<p>No candidates yet.</p>";
}
function renderVoters(){
```

```
 document.getElementById("voterTable").innerHTML=db.voters.map(v=>`<tr>
  <td>${escapeHTML(v.id)}</td><td>${v.voted?'<span class="badge">Voted</span>':'Not voted'}</td>
  <td><button class="btn secondary" onclick="removeVoter('${encodeURIComponent(v.id)}')">Remove</button></td></tr>`).join("")||'<tr><td colspan="3">No registered voter IDs imported.</td></tr>';
}
function renderAll(){
 document.getElementById("voterCount").textContent=db.voters.length;
 document.getElementById("voteCount").textContent=db.votes.length;

document.getElementById("turnout").textContent=(db.voters.length?Math.round(db.votes.length/db.voters.length*100):0)+"%";

document.getElementById("electionStatus").textContent=db.open?"OPEN":"CLOSED";
 document.getElementById("electionDescription").textContent=db.open?"Voting is open in this local demonstration.":"Voting is currently closed by the administrator.";
 document.getElementById("electionToggle").value=db.open?"open":"closed";
 renderCandidates();renderResults();renderVoters();
 if(!db.open)document.getElementById("submitVote").disabled=true;
}
function selectCandidate(id){
 if(!verified||!db.open)return;
 selectedCandidate=id;renderCandidates();
 const c=db.candidates.find(x=>x.id===id);
 document.getElementById("selectedMessage").innerHTML=`You selected <b>${escapeHTML(c.name)}</b> (${escapeHTML(c.party)}). Review before submitting.`;
 document.getElementById("submitVote").disabled=false;
```

```
}

function voterRecord(id){return
db.voters.find(v=>v.id.toLowerCase()===id.toLowerCase())}

function hasVoted(id){return
db.votes.some(v=>v.voterId.toLowerCase()===id.toLowerCase())}

document.getElementById("requestOtp").addEventListener("click",()=>{

 const id=document.getElementById("voterId").value.trim();

 const msg=document.getElementById("loginMessage");

 if(!id){msg.innerHTML='<div class="notice warning">Enter a voter ID
first.</div>';return}

 if(!db.open){msg.innerHTML='<div class="notice warning">This election is
closed.</div>';return}

 if(hasVoted(id)){msg.innerHTML='<div class="notice warning">This voter ID has
already voted.</div>';return}

 currentVoter=id;verified=false;selectedCandidate=null;

 pendingOtp=String(Math.floor(100000+Math.random()*900000));

 document.getElementById("demoOtp").textContent=pendingOtp;

 document.getElementById("otpInput").value="";

 document.getElementById("otpArea").classList.remove("hidden");

 document.getElementById("ballotArea").classList.add("hidden");

 msg.innerHTML='<div class="notice">OTP generated for this demo. It is not sent
by SMS or email.</div>';

});

document.getElementById("verifyOtp").addEventListener("click",()=>{

 const input=document.getElementById("otpInput").value.trim();

 if(!currentVoter||input!==pendingOtp){

  document.getElementById("loginMessage").innerHTML='<div class="notice
warning">Incorrect OTP. Please try again.</div>';return;

 }
```

```
if(hasVoted(currentVoter)){document.getElementById("loginMessage").innerHTML='<div class="notice warning">This voter has already voted.</div>';return}
 if(!db.voters.some(v=>v.id.toLowerCase()===currentVoter.toLowerCase())){
  db.voters.push({id:currentVoter,voted:false});
 }
 verified=true;pendingOtp="";
 document.getElementById("loginMessage").innerHTML='<div class="notice success">✓ Verification successful. Your ballot is unlocked.</div>';
 document.getElementById("ballotArea").classList.remove("hidden");
 document.getElementById("submitVote").disabled=true;
 renderAll();
});
document.getElementById("submitVote").addEventListener("click",()=>{
 if(!verified||!selectedCandidate||!db.open)return;
 if(hasVoted(currentVoter)){toast("This voter ID has already voted.");return}
 const c=db.candidates.find(x=>x.id===selectedCandidate);
 if(!c){toast("Candidate no longer exists.");return}
 if(!confirm("Submit your vote? This action cannot be undone in the demo."))return;
 db.votes.push({voterId:currentVoter,candidateId:c.id,submittedAt:new Date().toISOString()});
 const voter=voterRecord(currentVoter);if(voter)voter.voted=true;
 verified=false;pendingOtp="";selectedCandidate=null;
 document.getElementById("ballotArea").classList.add("hidden");
 document.getElementById("otpArea").classList.add("hidden");
 document.getElementById("loginMessage").innerHTML='<div class="notice success">✓ Vote recorded! Your confirmation ID is '+escapeHTML("VO-"+Date.now().toString().slice(-8))+'</div>';
 saveDB();toast("Vote submitted successfully.");showPage("history");
 document.getElementById("historyId").value=currentVoter;checkHistory();
```

```
});
function checkHistory(){
 const id=document.getElementById("historyId").value.trim();
 const result=document.getElementById("historyResult");
 if(!id){result.textContent="Enter a voter ID.";return}
 result.innerHTML=hasVoted(id)?'<span class="badge">Vote
submitted</span><br>Your demo vote has been recorded. Candidate choice is not
displayed.':
  'No vote has been recorded for this voter ID.';
}
function loginAdmin(){
 if(document.getElementById("adminPassword").value==="admin123"){

adminUnlocked=true;document.getElementById("adminLogin").classList.add("hid
den");

document.getElementById("adminArea").classList.remove("hidden");toast("Admin
console unlocked.");
 }else document.getElementById("adminMessage").innerHTML='<div
class="notice warning">Incorrect demo password.</div>';
}
function logoutAdmin(){

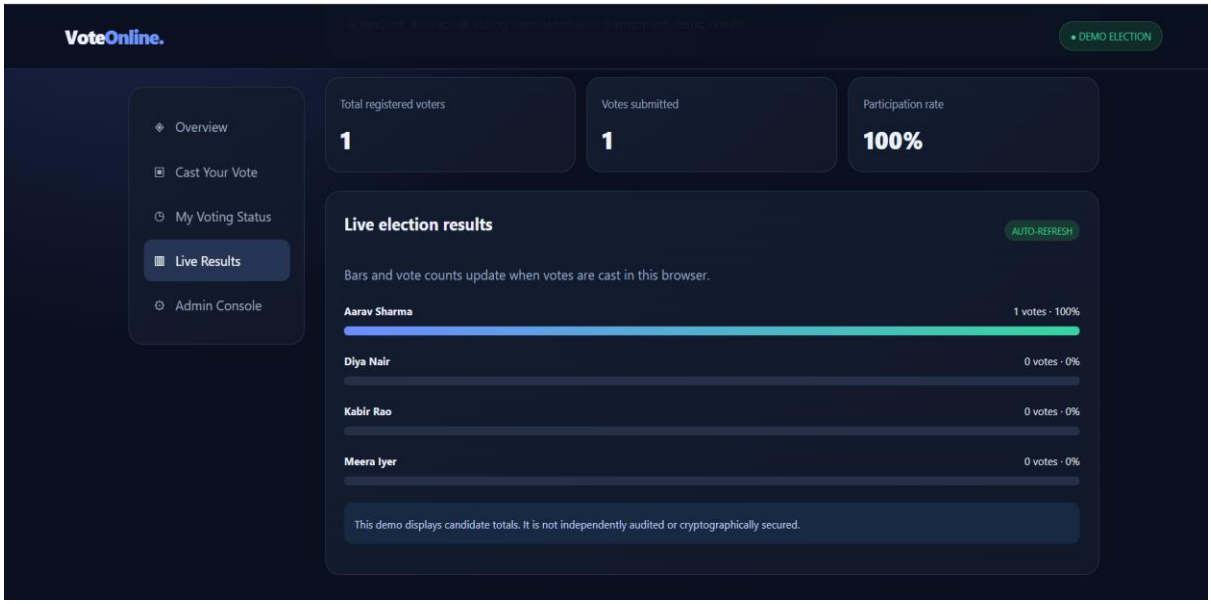
adminUnlocked=false;document.getElementById("adminLogin").classList.remove(
"hidden");

document.getElementById("adminArea").classList.add("hidden");document.getEle
mentById("adminPassword").value="";
}
function addCandidate(){
 if(!adminUnlocked)return;
```

```
 const name=document.getElementById("candidateName").value.trim();
 const party=document.getElementById("candidateParty").value.trim();
 const description=document.getElementById("candidateDescription").value.trim();
 if(!name||!party){toast("Candidate name and party are required.");return}
 db.candidates.push({id:db.nextId++,name,party,description,icon:" 🗳️ "});
 document.getElementById("candidateName").value="";
 document.getElementById("candidateParty").value="";
 document.getElementById("candidateDescription").value="";
 saveDB();toast("Candidate added.");
}
async function importVoters(){
 if(!adminUnlocked)return;
 const file=document.getElementById("voterFile").files[0];
 if(!file){toast("Choose a CSV or text file first.");return}
 const text=await file.text();
 const ids=text.split(/\r?\n/).map(line=>line.split(",")[0].trim().replace(/^"|"$/g,""))
  .filter((id,i)=>id&&!(i===0&&/voter.?id/i.test(id)));
 let added=0;
ids.forEach(id=>{if(!voterRecord(id)){db.voters.push({id,voted:hasVoted(id)});added++}});
 saveDB();toast(`${added} voter ID(s) imported.`);
}
function removeVoter(encoded){
 if(!adminUnlocked)return;
 const id=decodeURIComponent(encoded);
 if(hasVoted(id)){toast("Cannot remove a voter who has already voted.");return}
 db.voters=db.voters.filter(v=>v.id!==id);saveDB();toast("Voter removed.");
```

```
}
function setElectionStatus(value){
 if(!adminUnlocked)return;
 db.open=value==="open";saveDB();toast(db.open?"Election opened.":"Election closed.");
}
function resetElection(){
 if(!adminUnlocked)return;
 if(!confirm("Clear ALL demo votes and registered voter IDs? This cannot be undone."))return;
 db.votes=[];db.voters=[];saveDB();toast("Demo election data reset.");
}
function exportResults(){
 if(!adminUnlocked)return;
 const rows=[["Candidate","Party","Votes"],...db.candidates.map(c=>[
  c.name,c.party,db.votes.filter(v=>v.candidateId===c.id).length
 ])];
 const csv=rows.map(r=>r.map(v=>'"'+String(v).replace(/"/g,'""')+'"').join(",")).join("\r\n");
 const url=URL.createObjectURL(new Blob([csv],{type:"text/csv"}));
 const a=document.createElement("a");a.href=url;a.download="voteonline-results.csv";a.click();URL.revokeObjectURL(url);
}
function updateClock(){document.getElementById("clock").textContent=new Date().toLocaleTimeString()}
setInterval(updateClock,1000);updateClock();renderAll();
</script>
</body>
</html>
```

Output:



Result:

Thus, the VoteOnline e-voting system was developed successfully with interactive voter verification, candidate selection, vote submission, administrator management, and dynamically updated election results.